

MAT 115E Introduction to Programming Language

Lab-10/ CRN : 23487

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1 Question 1

We want to write a **C function** that searches for an element in a sorted array in an ascending order. If the element we are looking for is available in the array, your function should return index of element; otherwise, it should return -1. Furthermore, the main function should print the number of iterations required to complete the search.

Searching process should be done according to the following scenario:

- Step 1. Middle element of the array given in the question should be found at first, i.e. $\text{middleIndex} = (\text{minIndex} + \text{maxIndex}) / 2$
- Step 2. It is checked whether the middle is equal to the element we are looking for. If yes, return index of it.
- Step 3. If the element we are looking for is less than middle, then only the left part of the array from 0 to $(\text{middleIndex} - 1)$ is taken into account. Otherwise, the right part $(\text{middleIndex} + 1)$ to $(n - 1)$ is taken into account.
(n : number of array elements)
- Step 4. The first step is repeated for the considered part of array.

Hint: Your function should have four parameters: Array in an ascending order, size of such an array, element that are searched and reference to variable of iteration numbers.

2 Question 2

Suppose you have a catalogue of automobiles such as in the following table:

ID	1002	2176	3201	4204	5512	6309	8841	9004
Color								
0-black	0	1	2	1	2	0	1	0
1-blue								
2-red								
Price, USD	24000	18000	36000	26000	35000	29000	31000	38000

Write a **C function** that will input an array for ID, an array for Color, an array for Price and prints the ID numbers of the black cars that are priced under 30000 USD.

3 Question 3

The (x, y) coordinates of an object (in meters) as a function of time t are given by

$$x(t) = 5t - 10 \quad y(t) = 25t^2 - 120t + 144$$

The distance of the object from the reference point (x_0, y_0) is given by $d = \sqrt{(x - x_0)^2 + (y - y_0)^2}$. Create a **C function** that takes reference point coordinates as parameters and uses a **while** loop to increment time starting from $t = 0$ in steps of 0.01 seconds. At each step, compute the distance from the reference point and determine the first time t at which the distance becomes less than $d = 15$ meters. The function should return the last time step immediately before the distance drops below 15 meters, along with the corresponding distance at that time.